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Shallow Water Pipeline Installation Analysis

60 parametric S-lay cases: 2 vessels x 5 pipe sizes x 6 water depths

60 parametric cases analysed overnight

Generated 2026-04-15 04:05 UTC · demo_04

Methodology

This analysis screens shallow water S-lay pipeline installations across a parametric matrix of vessels, pipe sizes, and water depths. Each case solves the catenary mechanics to find the minimum horizontal tension and evaluates five installation checks per DNV-ST-F101 and API RP 1111 methodology.

S-lay Catenary Theory

During S-lay, the pipe forms a catenary curve from the vessel stinger tip down to the seabed touchdown point (TDP). The catenary is characterised by parameter $a = H / w_{sub}$, where H is horizontal tension and w_{sub} is submerged weight per unit length.

- Arc length: $s = a \times \sinh(x/a)$
- Height above TDP: $y = a \times (\cosh(x/a) - 1)$
- Pipe angle: $\theta = \arctan(\sinh(x/a))$

Check 1: Stinger Departure Angle

The required departure angle at the stinger tip must fall within the vessel's adjustable range. Large PLV: 2°-8°, Barge: 3°-12°.

Check 2: Top Tension

Tension at stinger departure: $T_{top} = H \times \cosh(x_{dep}/a)$. Must not exceed vessel tensioner capacity (600 te or 250 te).

Check 3: Sagbend Stress (Critical for Shallow Water)

At the TDP, curvature $\kappa = w_{sub} / H$ gives the tightest bend radius. Combined bending + axial stress: $\sigma = E \times OD / (2R) + H / A_{steel}$. Must satisfy $\sigma \leq 0.72 \times SMYS$ per DNV-ST-F101 installation allowable.

Check 4: Overbend Stress

Pipe bending over the stinger: $R_{ob} = L_{stinger} / \sin(\theta_{dep})$. Overbend stress must also satisfy the $0.72 \times SMYS$ limit.

Check 5: Vessel Capability

Pipe diameter must be within vessel's handling range and water depth within operational envelope.

Go/No-Go Criteria

- GO:** All checks pass AND max utilisation < 0.85
- MARGINAL:** All checks pass AND max utilisation in [0.85, 1.00]
- NO-GO:** Any check fails (utilisation > 1.00)

Applicable Codes & Standards

- DNV-ST-F101 (2021) - Submarine Pipeline Systems
- API RP 1111 (2015) - Design, Construction, Operation, and Maintenance of Offshore Hydrocarbon Pipelines
- DNV-OS-F101 (2013) - Submarine Pipeline Systems

Installation Screening Summary

All 60 parametric cases — 2 vessels x 5 pipe sizes x 6 water depths

VESSEL	PIPE	WT (MM)	DEPTH (M)	T_TOP (TE)	DEPART. ANGLE	SAGBEND UTIL	MAX UTIL.	GOV. CHECK	STATUS
Large PLV	8"	8.18	7	2.7	8.0 deg	10%	FAIL	vessel_capability	NO-GO
Large PLV	8"	8.18	10	3.9	8.0 deg	7%	7.0%	overbend	GO
Large PLV	8"	8.18	15	5.8	8.0 deg	5%	7.0%	overbend	GO
Large PLV	8"	8.18	20	7.7	8.0 deg	3%	7.0%	overbend	GO
Large PLV	8"	8.18	25	9.6	8.0 deg	3%	7.0%	overbend	GO
Large PLV	8"	8.18	30	11.5	8.0 deg	2%	7.0%	overbend	GO
Large PLV	12"	9.53	7	59.9	8.0 deg	15%	FAIL	vessel_capability	NO-GO
Large PLV	12"	9.53	10	85.6	8.0 deg	10%	14.3%	tension	GO
Large PLV	12"	9.53	15	128.4	8.0 deg	7%	21.4%	tension	GO
Large PLV	12"	9.53	20	171.2	8.0 deg	5%	28.5%	tension	GO
Large PLV	12"	9.53	25	214.0	8.0 deg	4%	35.7%	tension	GO
Large PLV	12"	9.53	30	256.7	8.0 deg	3%	42.8%	tension	GO
Large PLV	16"	9.53	7	77.3	8.0 deg	18%	FAIL	vessel_capability	NO-GO
Large PLV	16"	9.53	10	110.4	8.0 deg	13%	18.4%	tension	GO
Large PLV	16"	9.53	15	165.6	8.0 deg	9%	27.6%	tension	GO
Large PLV	16"	9.53	20	220.8	8.0 deg	6%	36.8%	tension	GO
Large PLV	16"	9.53	25	276.0	8.0 deg	5%	46.0%	tension	GO
Large PLV	16"	9.53	30	331.2	8.0 deg	4%	55.2%	tension	GO
Large PLV	20"	9.53	7	92.3	8.0 deg	23%	FAIL	vessel_capability	NO-GO
Large PLV	20"	9.53	10	131.9	8.0 deg	16%	22.0%	tension	GO
Large PLV	20"	9.53	15	197.8	8.0 deg	11%	33.0%	tension	GO
Large PLV	20"	9.53	20	263.7	8.0 deg	8%	44.0%	tension	GO
Large PLV	20"	9.53	25	329.7	8.0 deg	6%	54.9%	tension	GO
Large PLV	20"	9.53	30	395.6	8.0 deg	5%	65.9%	tension	GO
Large PLV	24"	9.53	7	122.9	8.0 deg	27%	FAIL	vessel_capability	NO-GO
Large PLV	24"	9.53	10	175.6	8.0 deg	19%	29.3%	tension	GO
Large PLV	24"	9.53	15	263.4	8.0 deg	13%	43.9%	tension	GO
Large PLV	24"	9.53	20	351.3	8.0 deg	10%	58.5%	tension	GO
Large PLV	24"	9.53	25	439.1	8.0 deg	8%	73.2%	tension	GO
Large PLV	24"	9.53	30	526.9	8.0 deg	6%	87.8%	tension	MARGINAL
Shallow Water Barge	8"	8.18	7	1.2	12.0 deg	22%	22.4%	sagbend	GO
Shallow Water Barge	8"	8.18	10	1.7	12.0 deg	16%	18.3%	overbend	GO

VESSEL	PIPE	WT (MM)	DEPTH (M)	T_TOP (TE)	DEPART. ANGLE	SAGBEND UTIL	MAX UTIL.	GOV. CHECK	STATUS
Shallow Water Barge	8"	8.18	15	2.6	12.0 deg	10%	18.3%	overbend	GO
Shallow Water Barge	8"	8.18	20	3.4	12.0 deg	8%	18.3%	overbend	GO
Shallow Water Barge	8"	8.18	25	4.3	12.0 deg	6%	18.3%	overbend	GO
Shallow Water Barge	8"	8.18	30	5.1	12.0 deg	5%	18.3%	overbend	GO
Shallow Water Barge	12"	9.53	7	26.7	12.0 deg	33%	33.2%	sagbend	GO
Shallow Water Barge	12"	9.53	10	38.1	12.0 deg	23%	27.0%	overbend	GO
Shallow Water Barge	12"	9.53	15	57.2	12.0 deg	15%	27.0%	overbend	GO
Shallow Water Barge	12"	9.53	20	76.2	12.0 deg	12%	30.5%	tension	GO
Shallow Water Barge	12"	9.53	25	95.3	12.0 deg	9%	38.1%	tension	GO
Shallow Water Barge	12"	9.53	30	114.3	12.0 deg	8%	45.7%	tension	GO
Shallow Water Barge	16"	9.53	7	34.4	12.0 deg	42%	41.6%	sagbend	GO
Shallow Water Barge	16"	9.53	10	49.2	12.0 deg	29%	33.9%	overbend	GO
Shallow Water Barge	16"	9.53	15	73.8	12.0 deg	19%	33.9%	overbend	GO
Shallow Water Barge	16"	9.53	20	98.3	12.0 deg	15%	39.3%	tension	GO
Shallow Water Barge	16"	9.53	25	122.9	12.0 deg	12%	49.2%	tension	GO
Shallow Water Barge	16"	9.53	30	147.5	12.0 deg	10%	59.0%	tension	GO
Shallow Water Barge	20"	9.53	7	41.1	12.0 deg	52%	52.0%	sagbend	GO
Shallow Water Barge	20"	9.53	10	58.7	12.0 deg	36%	42.4%	overbend	GO
Shallow Water Barge	20"	9.53	15	88.1	12.0 deg	24%	42.4%	overbend	GO
Shallow Water Barge	20"	9.53	20	117.5	12.0 deg	18%	47.0%	tension	GO
Shallow Water Barge	20"	9.53	25	146.8	12.0 deg	15%	58.7%	tension	GO
Shallow Water Barge	20"	9.53	30	176.2	12.0 deg	12%	70.5%	tension	GO

VESSEL	PIPE	WT (MM)	DEPTH (M)	T_TOP (TE)	DEPART. ANGLE	SAGBEND UTIL	MAX UTIL.	GOV. CHECK	STATUS
Shallow Water Barge	24"	9.53	7	54.8	12.0 deg	62%	62.4%	sagbend	GO
Shallow Water Barge	24"	9.53	10	78.2	12.0 deg	44%	50.8%	overbend	GO
Shallow Water Barge	24"	9.53	15	117.3	12.0 deg	29%	50.8%	overbend	GO
Shallow Water Barge	24"	9.53	20	156.4	12.0 deg	22%	62.6%	tension	GO
Shallow Water Barge	24"	9.53	25	195.5	12.0 deg	17%	78.2%	tension	GO
Shallow Water Barge	24"	9.53	30	234.7	12.0 deg	15%	93.9%	tension	MARGINAL

Take This Analysis Live During Operations

LIVE

This report used **design sea states** to screen parametric cases overnight. During the actual operation, **digitalmodel** can feed your vessel's measured motion data — VMMS, IMMS, MRU — directly into the same engineering models for **real-time go/no-go decisions**.

Instead of relying on forecasted Hs limits, the pipeline installation screening updates continuously with actual crane tip motions, hook loads, and dynamic amplification factors measured on your vessel.

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Assumptions & Limitations

- All pipes are X65 grade (SMYS = 448 MPa, E = 207 GPa)
- Catenary solution assumes no current loading on the suspended span
- Sagbend and overbend stresses use simplified elastic beam theory
- Combined stress is conservative additive (bending + axial) per DNV-ST-F101 Sec 5
- Stinger radius approximated as $R_{ob} = L_{stinger} / \sin(\theta_{dep})$
- No dynamic amplification applied — quasi-static installation assumed
- Pipe submerged weight includes anti-corrosion coating and CWC where applicable
- Seabed assumed flat at each water depth — no slope corrections
- Seawater density = 1025 kg/m3 throughout water column
- Installation stress limit factor = 0.72 per DNV-ST-F101 (2021)

Chart 1: Go/No-Go Installation Matrix

Side-by-side vessel comparison across all pipe sizes and water depths.

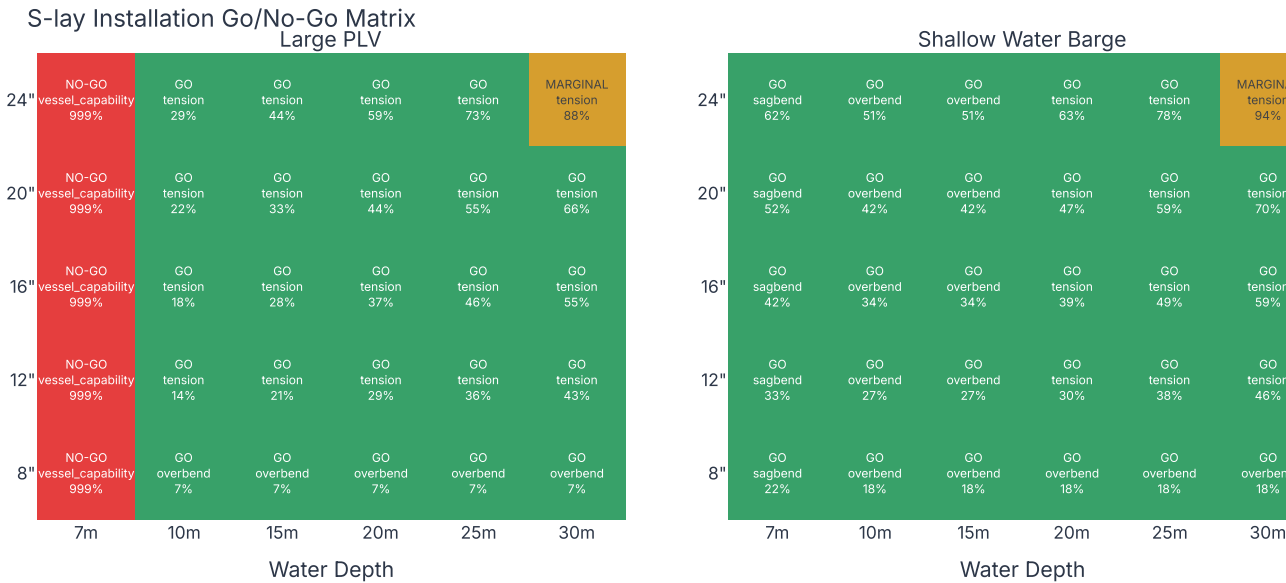


Chart 2: Sagbend Stress Utilisation Heatmap

Sagbend utilisation as percentage of 0.72 x SMYS (X65 = 448 MPa). Green = low, red = critical.

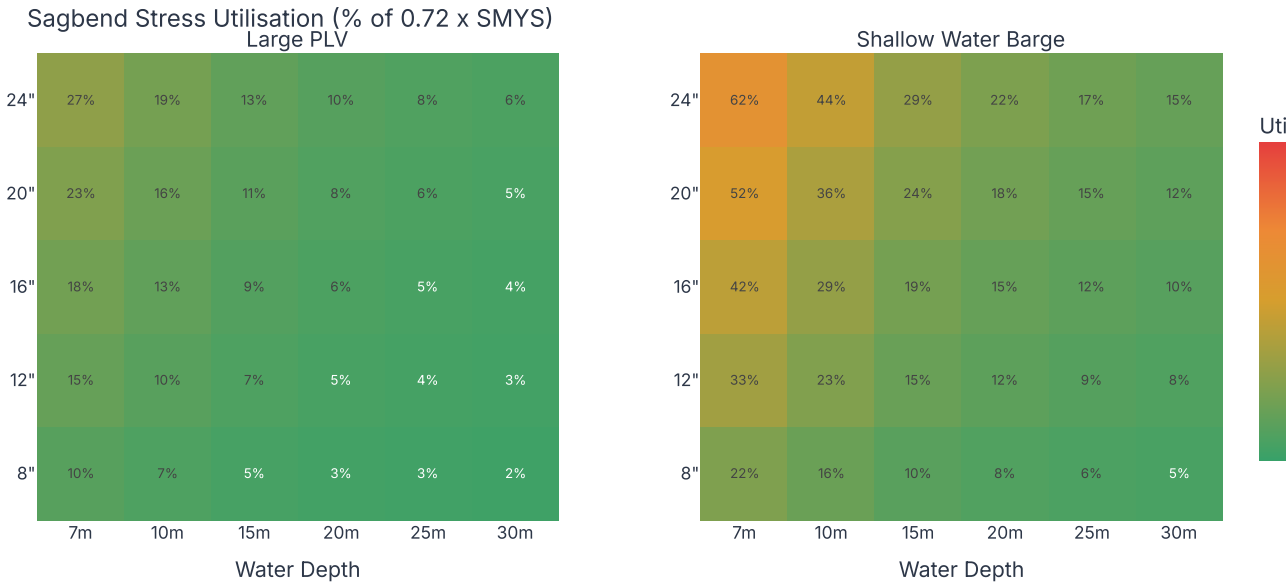


Chart 3: Required Top Tension vs Water Depth

Solid = Large PLV, dashed = Barge. Horizontal lines show tensioner capacity limits.

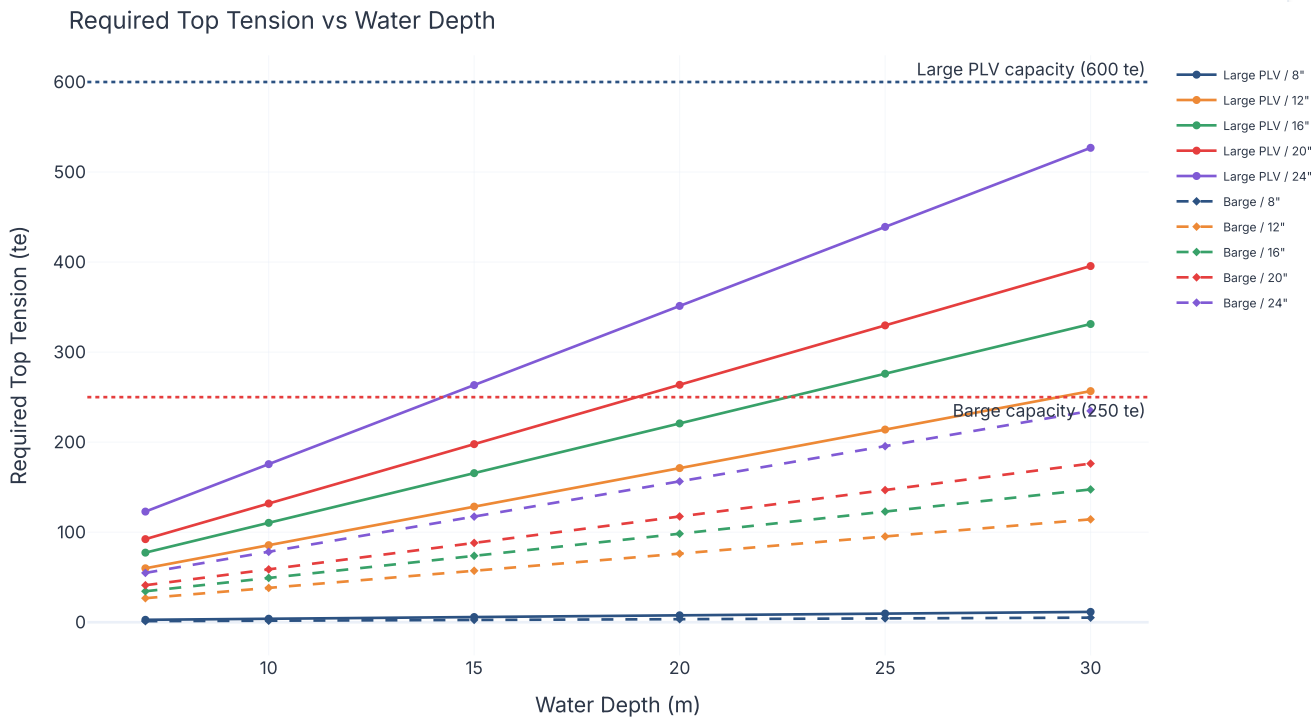


Chart 4: Stinger Departure Angle vs Water Depth

Required departure angle with vessel stinger range bands. Shows where stinger geometry governs.

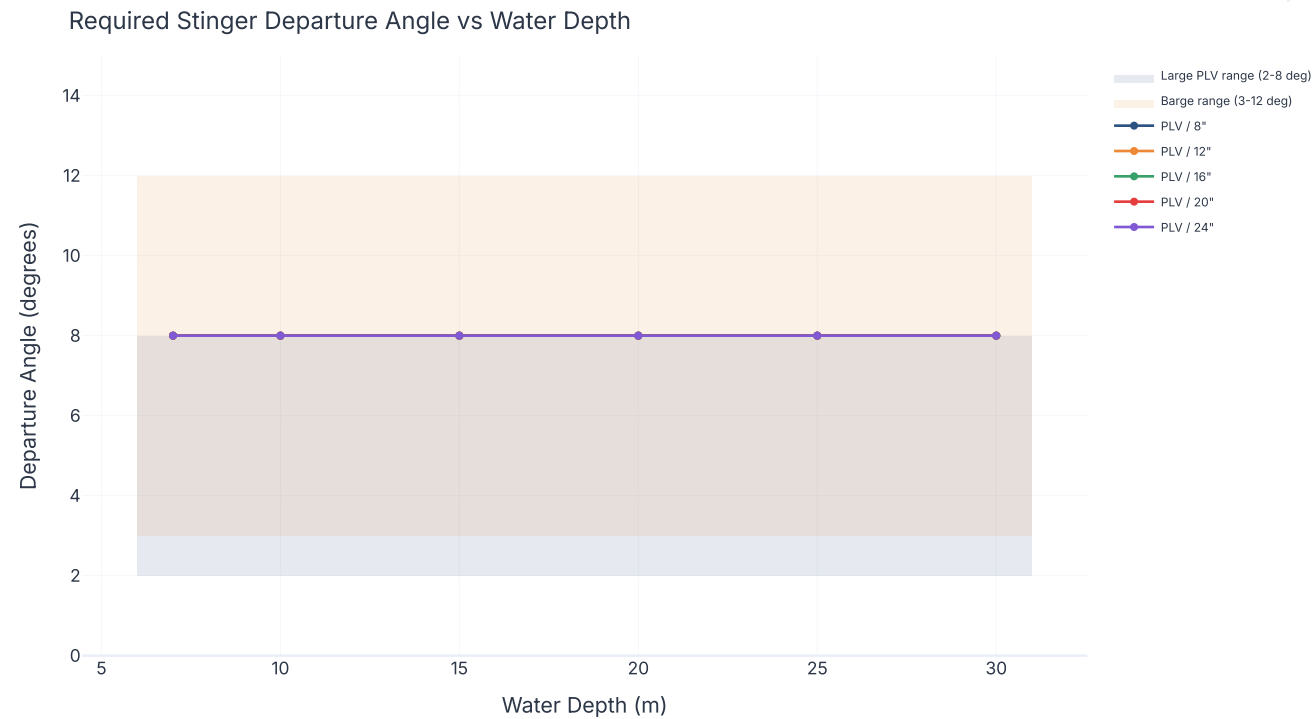
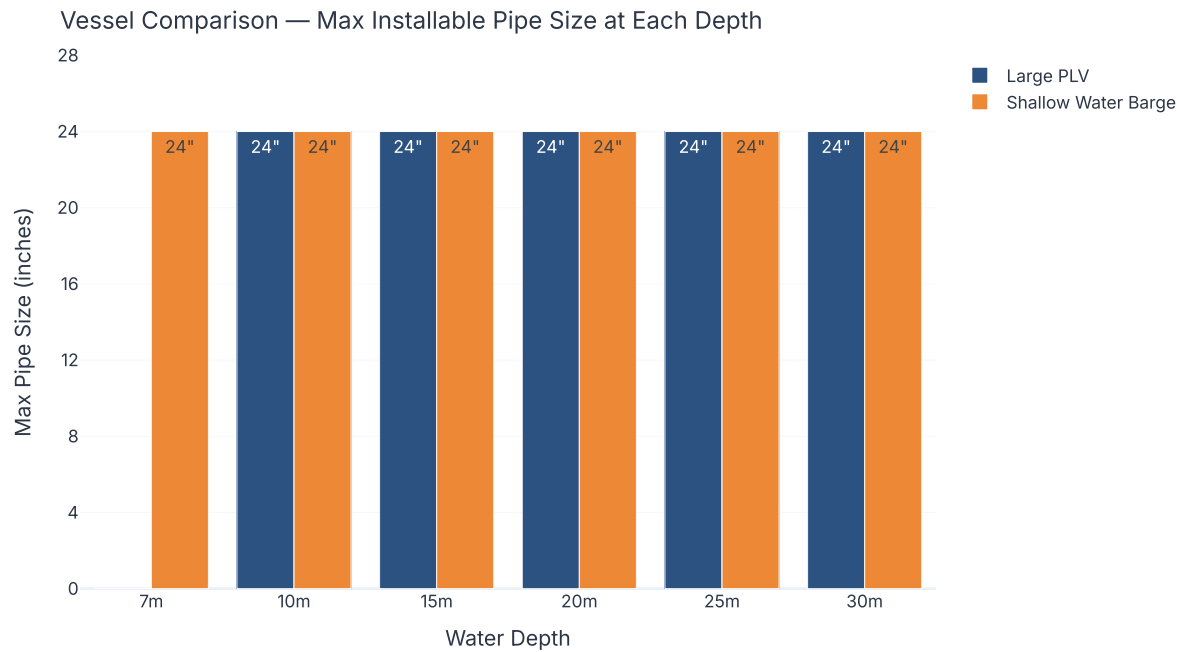


Chart 5: Vessel Head-to-Head — Max Pipe Size at Each Depth

Maximum installable pipe diameter for each vessel at each water depth.



These results are preliminary engineering estimates generated by automated parametric analysis. All outputs require review by a qualified engineer before use in design or operational decisions.

References: DNV-ST-F101 (2021) - Submarine Pipeline Systems · API RP 1111 (2015) - Design, Construction, Operation, and Maintenance of Offshore Hydrocarbon Pipelines · DNV-OS-F101 (2013) - Submarine Pipeline Systems
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